

FOUNDATIONS OF COMPUTER SCIENCE

College exam notes | Units 1 and 2

A consolidated study guide covering computer fundamentals, architecture, functional units, buses, memory hierarchy, RAM and ROM, and secondary storage. Practice-question and answer-key slides are omitted.

Study map

Unit 1 - Introduction to Computers	Unit 2 - Memory Hierarchy and Classification
Computer basics and characteristics; everyday uses; computer categories; early milestones and generations; Von Neumann and Harvard architectures; CPU registers; input, output, memory and buses.	Memory cells and units; hierarchy; volatile and non-volatile memory; SRAM, DRAM and ROM; HDD, SSD and flash media; removable and optical storage.

The notes merge overlapping slides from the supplied versions. Generation dates are approximate because the presentations use slightly different foundations.

UNIT 1 | INTRODUCTION TO COMPUTERS

1. Computer basics

A computer is an electronic machine that accepts data as input, processes it according to instructions in a program, and produces information as output. It can also store data and results for later use. Examples of tasks include calculations, data analysis, file management, graphic design and internet access.

Term	Meaning	Examples
Data	Raw facts or values entered for processing.	Numbers, text, images, sensor readings.
Information	Processed data presented in a useful form.	A calculated total, report, chart or displayed result.
Hardware	The physical parts of a computer system.	CPU, monitor, keyboard, mouse, storage devices.
Software	Programs and instructions that tell hardware what to do.	Operating systems, applications, utilities.

Basic operation: Input data -> Process it using a program -> Output information. Memory and storage hold instructions, current data, intermediate results and saved files.

2. Characteristics, advantages and limitations

Characteristic	Exam-ready meaning
Speed	Processes large amounts of data quickly. Processing speed may be expressed in microseconds (10^{-6} s), nanoseconds (10^{-9} s) or picoseconds (10^{-12} s).
Accuracy	Produces precise results when the data and program are correct; incorrect input or instructions can still produce incorrect output.
Automation	After receiving instructions and data, carries out the programmed steps with little ongoing human intervention.
Diligence	Can repeat tasks for long periods without fatigue or loss of concentration.
Versatility	Can perform many different kinds of work and run tasks concurrently.
Storage and recall	Stores large amounts of data and retrieves saved information when needed.
No feelings	Does not have human emotions or personal judgment; it follows programs and input.

Advantages	Limitations
Speed and efficiency: completes repetitive work quickly and saves time. Accuracy: reduces calculation errors when given correct input and programs. Storage: keeps large amounts of organized, accessible data. Multitasking and versatility: supports different tasks and fields.	Health issues: prolonged use can cause eye strain, back pain and repetitive strain. Cost: setup, maintenance and updates can be expensive. Dependency: over-reliance can weaken critical thinking and manual skills. Security risks: hacking, malware and data breaches. Technical issues: hardware or software failure can interrupt work.

3. Computer applications in daily life

Area	Common uses from the presentations
Education	Online classes and e-learning; research and digital libraries; simulations and virtual labs; student records, result processing, question preparation, handouts and notes.
Banking and finance	Online banking, fund transfers and bill payment; MICR for reading and sorting cheques; ATMs for deposits and withdrawals; digital payments and stock trading.
Healthcare	Electronic patient records; diagnostic data and lab reports; patient monitoring; CT, MRI and ECG systems; pharmacy checks; computer-assisted and robotic surgery.
Engineering	Structural stress and strain analysis; industrial system design and simulation; city, building and apartment plans using 2D and 3D CAD.
Communication and work	Email, messaging and video calls; word processing, spreadsheets and presentations; databases, customer records, payroll and inventory systems.
Transport and navigation	Ride-hailing, GPS maps, traffic information and public-transport updates.
Home, entertainment and creative work	Smart-home controls and security; streaming, gaming and music; e-commerce and order tracking; photo editing, video, 3D design, writing and publishing.

4. Computer classifications

By size and power

Type	Main role and typical work
Microcomputer / personal computer	Small, affordable system for one person. Includes desktop, laptop, tablet and smartphone use for study, office work, browsing and gaming.
Minicomputer / mid-range system	Supports several users or departmental applications, such as business data and databases.
Mainframe	Large, reliable system for very high-volume transactions and records, such as banking, airline or railway reservations, insurance and government data.
Supercomputer	Very high computing performance for scientific calculations, weather and climate forecasting, space and nuclear research, simulation and AI.
Workstation	High-performance computer for one professional user, such as CAD, 3D graphics, engineering, research and software development.
Server	Provides shared services and resources over a network, including websites, applications, files, databases, email and cloud services.

By the kind of data processed

Type	How it works	Example or use
Analog	Represents continuously changing physical quantities.	Measurement, monitoring, simulation and control; aircraft or flight simulation.
Digital	Processes discrete values, represented in computers as binary 0s and 1s.	Modern laptops, phones and servers.
Hybrid	Combines analog measurement or speed with digital processing, storage or control.	Medical and industrial systems.

5. Early development and generations

Selected milestones

Milestone	Contribution
1642 - Blaise Pascal	Designed a mechanical adding machine.
1671 - Gottfried Wilhelm von Leibniz	Developed a calculator capable of multiplication.
1822 - Charles Babbage	Designed the Difference Engine. Babbage is often called the father of the modern digital computer.
1842 - Analytical Engine (date cited in the lecture slides)	Babbage designed an automatic machine intended to perform arithmetic operations. His work established principles used in later digital computer design.
Around 1880 - Herman Hollerith	Applied punched cards as an input medium; punched cards remained in use for decades. Keyboard machines also began to appear around this period.

Computer generations

Generation / period	Core technology	Languages, systems and main changes	Examples
First 1946-1959	Vacuum tubes.	Machine language; batch processing; punch cards, paper tape and magnetic tape. Very large, costly, unreliable, hot and power-hungry; often required air conditioning.	ENIAC, EDVAC, UNIVAC I, IBM 701.
Second 1959-1965	Transistors; magnetic-core primary memory, with magnetic tape and disks for secondary storage.	Assembly language and high-level FORTRAN/COBOL; batch processing and multiprogramming. Smaller, faster, more reliable and lower power than first generation, but still costly.	IBM 1401, IBM 7094, CDC 1604, UNIVAC 1108.
Third 1965-1971	Integrated circuits (ICs), combining many components on chips.	High-level languages such as FORTRAN, COBOL and PASCAL; multiprogramming, remote processing and time-sharing. More compact, reliable and efficient, with less heat and maintenance.	IBM 360 series, PDP-8, IBM 370 series.
Fourth about 1971-1980	VLSI and microprocessors.	CPU functions integrated onto a chip; personal computers became common; networking grew. High-level languages included C and C++; operating systems supported time-sharing, real-time and distributed work.	Apple II, IBM PC, PDP-11, DEC 10, Cray-1.
Fifth 1980 onward	ULSI, AI-focused software, parallel processing.	Machine learning, natural-language processing, more user-friendly multimedia interfaces and powerful compact systems; high-level languages include Python and Prolog.	Modern AI assistants and advanced supercomputers.

Generation dates are approximate. The presentations use some broader alternate date ranges, especially for the fourth generation. Identify each generation by its dominant hardware technology.

Evolution of computers	Computer generation
The whole history and continuous development of computing, including changes in technology, software and use.	One defined stage in that history, commonly identified by a major change in core hardware technology.

6. Computer architecture and CPU operation

Stored-program idea

A fixed-program computer performs a specific function and cannot readily be reprogrammed, such as a basic calculator. A stored-program computer keeps programs in memory so the same machine can perform many tasks. Von Neumann architecture, also called Princeton architecture, is a stored-program model credited to John von Neumann; the lecture deck dates it to 1945.

Feature	Von Neumann architecture	Harvard architecture
Memory	Instructions and data share one physical memory.	Instructions and data use separate memories.
Bus structure	Common bus carries instruction and data transfers.	Separate instruction and data buses.
Access	Shared path can require instruction and data access to take turns, limiting throughput.	Separate paths can allow instruction fetch and data access at the same time.
Typical trade-off and use	Simpler, lower-cost design; common in general-purpose computers.	More complex and costly; common in microcontrollers and digital signal processing.

Exact instruction timing depends on the processor design; the important distinction is the shared versus separate memory and bus paths.

Von Neumann strengths and uses	Main limitations
A unified memory makes the design simpler and cost-effective, supports flexible general-purpose programs and is easy to program. It underlies desktops, laptops and smartphones, and supports software development, education, gaming and multimedia.	Instructions and data share memory and a bus, so they compete for transfers. The resulting memory bottleneck and latency can slow sequential processing, limit throughput for highly parallel work and increase energy spent on repeated memory access.

Main CPU components and registers

Component or register	Role
Control Unit (CU)	Directs instruction execution and coordinates data flow among the CPU, memory and input/output devices. It controls operations rather than performing arithmetic itself.
Arithmetic Logic Unit (ALU)	Performs arithmetic such as addition, subtraction, multiplication and division, plus logical comparisons such as greater than, less than and equal to.
Program Counter (PC)	Holds the address of the next instruction to be fetched.
Instruction Register (IR)	Holds the current instruction being decoded or executed.
Memory Address Register (MAR)	Holds the address of the memory location being accessed.
Memory Data Register (MDR)	Temporarily holds data moving to or from memory.
Accumulator and general-purpose registers	Hold intermediate arithmetic/logic results and temporary data during processing.

Fetch-decode-execute cycle

Step	What happens
1. Fetch	The PC identifies the next instruction address. The CPU obtains that instruction from memory and places it in the IR.
2. Decode	The CU interprets the instruction and determines which operation and data are required.
3. Fetch operands	If the instruction needs data, the CPU obtains the operands from memory or registers.
4. Execute and store	The ALU or another CPU unit performs the operation. The result is placed in a register or written to memory.
5. Repeat	The PC advances or changes to the next instruction address, and the cycle continues.

7. Functional units and system buses

Unit	Purpose and examples
Input unit	Accepts data and instructions from the user or devices; converts them into a form the computer can process and passes them into the system. Examples: keyboard, mouse, scanner and microphone.
Memory / storage unit	Stores instructions and data before processing, holds intermediate and final results, and keeps saved data for later use. Primary memory is fast and temporary; secondary storage is larger and persistent.
CPU	Executes program instructions. The CU coordinates system operations and the ALU performs arithmetic and logical work; registers hold current values and instructions.
Output unit	Presents processed results in a form usable by people or other systems. Examples: monitor, printer, speaker and headphones.

A simplified flow is: Input -> Memory/CPU processing -> Output. Data buses carry values; address buses identify locations; control signals coordinate when and how transfers occur.

Bus	Direction and purpose	Key exam point
Data bus	Carries data between CPU, memory and input/output devices; generally bidirectional.	Width is the number of bits transferred at once. Common widths include 8, 16, 32 and 64 bits.
Address bus	Carries the selected memory or device address, usually from the CPU toward memory or I/O; unidirectional in the basic model.	An n-bit address bus can identify 2^n locations. If byte-addressable: 4 bits = 16 bytes; 16 bits = 64 KB; 20 bits = 1 MB, using 1 KB = 1,024 bytes.
Control bus	Carries command and coordination signals such as read, write, timing and status/response signals; signals can travel in both directions.	Indicates what operation should occur and helps synchronize components.

UNIT 2 | MEMORY HIERARCHY AND CLASSIFICATION

1. Memory fundamentals and units

Computer memory holds the data and instructions needed for processing and preserves results. A memory cell stores one bit. In many systems, a memory address identifies a byte. A word is a group of bits handled together by the processor.

Unit	Course convention
Bit (b)	Smallest unit; a binary value of 0 or 1.
Byte (B)	8 bits.
Kilobyte (KB)	1,024 bytes.
Megabyte (MB)	1,024 KB.
Gigabyte (GB)	1,024 MB.
Terabyte (TB)	1,024 GB.
Petabyte (PB), Exabyte (EB), Zettabyte (ZB), Yottabyte (YB)	Each successive unit is 1,024 of the preceding unit under the binary convention used in the slides.

When answering from these lecture slides, use the 1,024-based convention: 1 KB = 1,024 bytes. Memory capacity is the total amount of data that can be stored.

2. Memory hierarchy and classification

Memory hierarchy places the smallest, fastest storage closest to the CPU and larger, slower storage farther away. Typical order: registers -> cache -> main memory (RAM) -> secondary storage (SSD/HDD) -> optical discs or magnetic tape for removable or archival use.

Level	Where / role	Typical trade-off
Registers	Inside the CPU; hold values used immediately by instructions.	Fastest access, very small capacity, high cost per bit.
Cache (L1, L2, L3)	On or close to the CPU; keeps frequently used instructions and data so the CPU can avoid slower RAM access.	Very fast, small. L1 is closest; L2 is larger; L3 may be shared.
Main memory (RAM)	Working area for the operating system, active programs and current data.	Larger than cache, slower, usually volatile.
Secondary / auxiliary storage	Persistent storage such as SSDs, HDDs, flash drives, memory cards, optical discs and magnetic tape.	Much larger capacity and lower cost per bit, but slower than CPU memory.

Classification	Main properties	Examples
Primary / internal memory	Fast, directly used during CPU work; includes registers, cache and RAM. The lecture slides also group ROM with primary memory.	Registers, cache, RAM; ROM for firmware.
Secondary / auxiliary memory	Non-volatile storage, generally accessed through input/output channels; larger and slower than primary memory.	HDD, SSD, flash drives, optical discs, magnetic tape.
Fixed secondary storage	Built into the system and normally stays installed.	HDD and SSD.
Removable secondary storage	Can be inserted or removed and carried between compatible systems.	USB drives, SD cards, CDs/DVDs/Blu-ray discs, removable magnetic media.

Terminology check: Cache is a separate high-speed layer between CPU registers and RAM. Main memory usually means RAM. ROM is non-volatile firmware storage, even when a course diagram groups it under primary memory.

Volatile and non-volatile memory

Property	Volatile	Non-volatile
Power removed	Stored working data is lost.	Stored data remains.
Speed / role	Generally faster; holds data and programs currently being processed.	Generally slower; holds firmware or files that must persist.
Capacity and cost	Usually smaller and more expensive per unit of capacity.	Usually larger and less expensive per unit of capacity.
Examples	RAM (SRAM and DRAM).	ROM, SSD, HDD, flash drives, SD cards and optical media.

3. RAM: SRAM and DRAM

Random Access Memory (RAM) is working memory for active programs and data. RAM is volatile. The two key types differ in how they store bits.

Feature	SRAM - Static RAM	DRAM - Dynamic RAM
Storage mechanism	Uses transistor-based flip-flop circuits to hold each bit while power is on.	Uses capacitors to store bits; charge leaks, so data must be refreshed periodically.
Speed	Very fast; does not need periodic refresh.	Slower than SRAM because of refresh requirements.
Cost and density	More expensive per bit and lower density; used in smaller amounts.	Cheaper per bit and higher density; practical for larger capacities.
Main use	CPU cache (L1, L2 and L3); storage-controller buffers; microcontrollers and embedded systems in vehicles, industrial equipment and medical devices.	Main system memory in computers, phones, tablets and gaming systems; also TVs, cars, cameras, printers and graphics memory such as GDDR.
Volatility	Volatile.	Volatile.

Quick distinction: SRAM = flip-flops, no refresh, faster, costlier, cache. DRAM = capacitors, refresh needed, denser, cheaper, main memory.

4. ROM and firmware

Read Only Memory (ROM) is non-volatile. It stores persistent instructions such as firmware and startup code used to initialize the computer and begin loading the operating system.

Type	How it is programmed or erased	Typical point
MROM (Mask ROM)	Programmed during manufacture.	Fixed contents; useful when many devices need the same unchanging program.
PROM	User can program it once; it cannot then be erased and rewritten.	Programmable Read Only Memory.
EPROM	Erased with ultraviolet light and then reprogrammed.	Must be erased before writing new contents.
EEPROM	Erased and reprogrammed electrically; selected bytes may be changed.	Can be updated without UV light.
Flash memory	Electrically erased and written in blocks.	Non-volatile storage used in SSDs, USB drives and memory cards.

5. Hard disk drives and solid-state drives

How an HDD works

A Hard Disk Drive stores data magnetically on spinning platters. A read/write head moves across the platter surface to access data. A spindle rotates the platters; an actuator arm and actuator position the head. The case, controller board, connectors and power connection support the drive.

HDD terms shown in the lecture diagram include platter, spindle, slider/read-write head, actuator arm, actuator axis, actuator, controller/circuit board, power and data connectors, jumper and casing.

How an SSD works

A Solid-State Drive stores data electronically in NAND flash memory and has no moving mechanical parts. It provides faster access and boot times than a mechanical HDD and is more resistant to shock during movement. Flash cells have a limited number of program/erase cycles.

Feature	HDD	SSD
Technology	Magnetic platters and moving read/write parts.	NAND flash memory and electronic controller; no moving parts.
Speed	Slower boot, application loading and random data access.	Faster boot and file access; NVMe drives can be much faster than SATA drives.
Durability and noise	More vulnerable to shock because of moving parts; can make mechanical noise.	More shock-resistant in normal use and silent.
Power	Generally uses more power.	Generally uses less power, which can help battery life.
Capacity and cost	Usually lower cost per GB and useful for large, economical storage.	Usually higher cost per GB; preferred when speed and responsiveness matter.
Wear and fragmentation	Mechanical parts wear; fragmentation can affect performance.	Flash write cycles are limited; fragmentation usually has little practical effect on performance.
Common uses	Backups, surveillance recordings and large media collections.	Operating systems, applications, games and performance-sensitive work.

SSD form factors and interfaces

Form / interface	Description and approximate slide speeds
2.5-inch SATA SSD	Older, common form factor; compatible with many desktop and laptop systems. SATA III is about 500-550 MB/s.
M.2 SATA SSD	Smaller M.2 form, but still limited by SATA speed, about 550 MB/s.
M.2 NVMe SSD	Uses PCIe through an M.2 slot; common in newer systems. Approximate values in the slides: PCIe 3.0 about 2,000-3,500 MB/s; PCIe 4.0 about 5,000-7,500 MB/s; PCIe 5.0 can exceed 10,000 MB/s.
PCIe add-in card	Installs in a PCIe slot; high performance, often used in workstations and servers; larger than an M.2 drive.
External SSD	Connects over USB 3.2, USB 4 or Thunderbolt; speed depends on the connection and drive. Useful for transfer and backup.

The listed speeds are approximate interface examples from the slides. Actual performance depends on the drive, controller, PCIe generation, connection and workload.

Capacity examples from the slides: 250-500 GB for an operating system and a few applications; around 1 TB as a general-purpose balance for software and games; 2 TB or more for large game libraries or video work.

Storage choice: SSD for fast booting and application response; HDD for high capacity at lower cost per GB.

6. Flash media, SD cards and USB drives

Flash media is portable, non-volatile electronic storage based on flash memory, commonly NAND. Data remains when power is removed. Its lack of moving mechanical parts makes it compact, quiet and more resistant to shock than an HDD. Examples include SD and microSD cards, USB flash drives and CompactFlash cards.

SD cards

An SD (Secure Digital) card is removable flash storage. Standard SD cards are common in cameras and larger devices; microSD cards fit smaller devices such as supported phones, tablets, drones and action cameras. Cards vary in capacity and speed for different uses. Applications include photos, high-resolution video, games, CCTV recording, maps, configuration files and embedded software.

USB flash drives

A USB flash drive, pen drive or USB memory stick is a small removable drive that connects through a USB port. It stores and transfers documents, photos, videos and other files. Common uses include file transfer, an extra backup copy, carrying study or work files, and preparing bootable media for operating-system installation or repair. USB drives range from a few gigabytes to hundreds of gigabytes or more.

Flash memory writes data into electronic cells and reads stored values back to the device. It retains information when power is removed.

Feature	SD card	USB flash drive
Common purpose	Storage inside compatible cameras, phones, drones and other portable devices.	File transfer and portable storage with computers and laptops.
Connection	SD or microSD slot.	USB port.
Size and portability	Very small; highly portable.	Small and portable.
Technology and retention	NAND flash; no moving parts; non-volatile.	NAND flash; no moving parts; non-volatile.

Advantages and limitations of flash media

Advantages	Limitations
Non-volatile; portable and compact; quick access; no moving parts; generally shock-resistant and energy-efficient; easy to connect or remove.	Limited program/erase cycles; quality and durability vary; small cards and drives can be lost or damaged; controller or cell failure can make data inaccessible. Keep another backup of important files.

7. Optical storage

Medium	How it stores data / capacity	Types and key facts
Compact Disc (CD)	Optical secondary storage on a circular polycarbonate disc, organized in tracks and sectors. Capacity up to about 700 MB.	CD-ROM is read-only. CD-R can be recorded once. CD-RW can be erased and rewritten.
Blu-ray Disc	Optical disc read with a blue-violet laser; capacity in the lecture slides is up to 25 GB.	Requires a Blu-ray reader. Higher capacity than a CD. The name combines "blue" (the laser) and "ray" (optical ray).

8. High-yield distinctions

Compare	Core distinction
CU vs ALU	CU directs and coordinates; ALU performs arithmetic and logical operations.
Data vs address bus	Data bus carries values and is bidirectional; address bus selects a location and is unidirectional in the basic CPU-to-memory model.
RAM vs ROM	RAM is volatile working memory; ROM is non-volatile and stores firmware.
SRAM vs DRAM	SRAM uses flip-flops and needs no refresh; DRAM uses capacitors and must be refreshed.
HDD vs SSD	HDD uses spinning magnetic platters; SSD uses flash memory with no moving parts.
SD vs USB flash drive	SD cards fit SD/microSD slots in portable devices; USB drives plug into a USB port and commonly transfer files between computers.
Von Neumann vs Harvard	Von Neumann shares instruction/data memory and bus; Harvard separates them and can access both paths at once.

End of notes